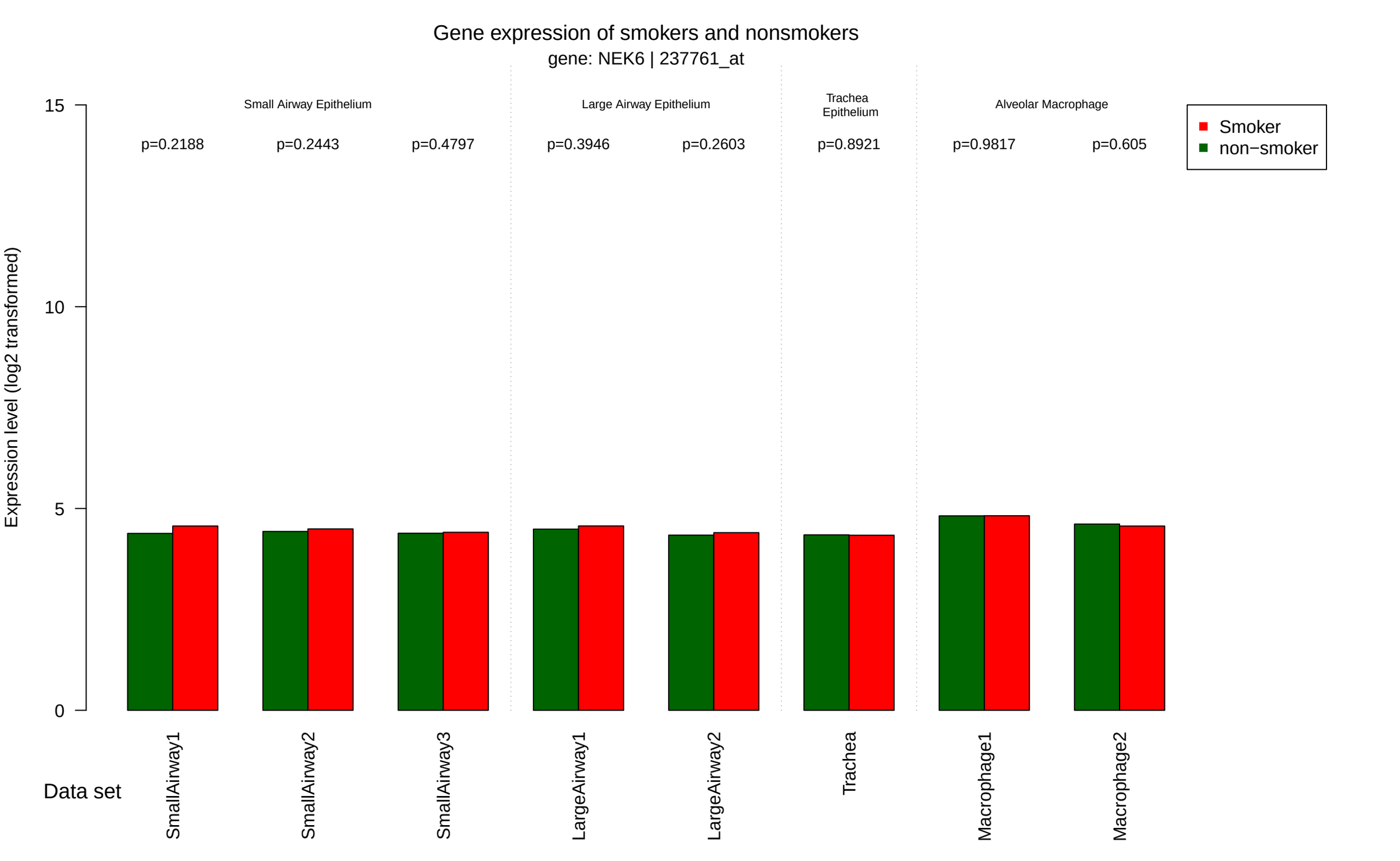
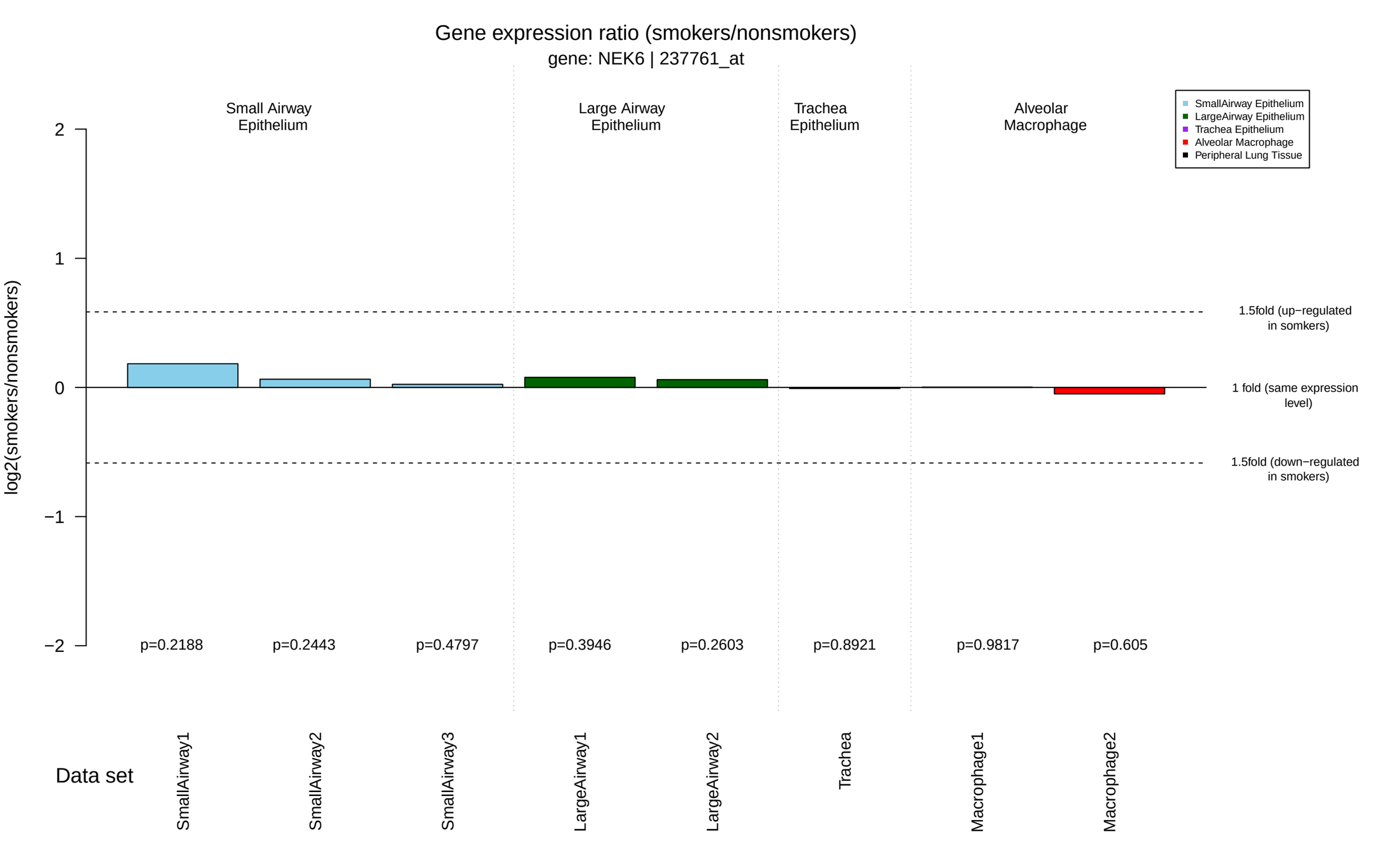
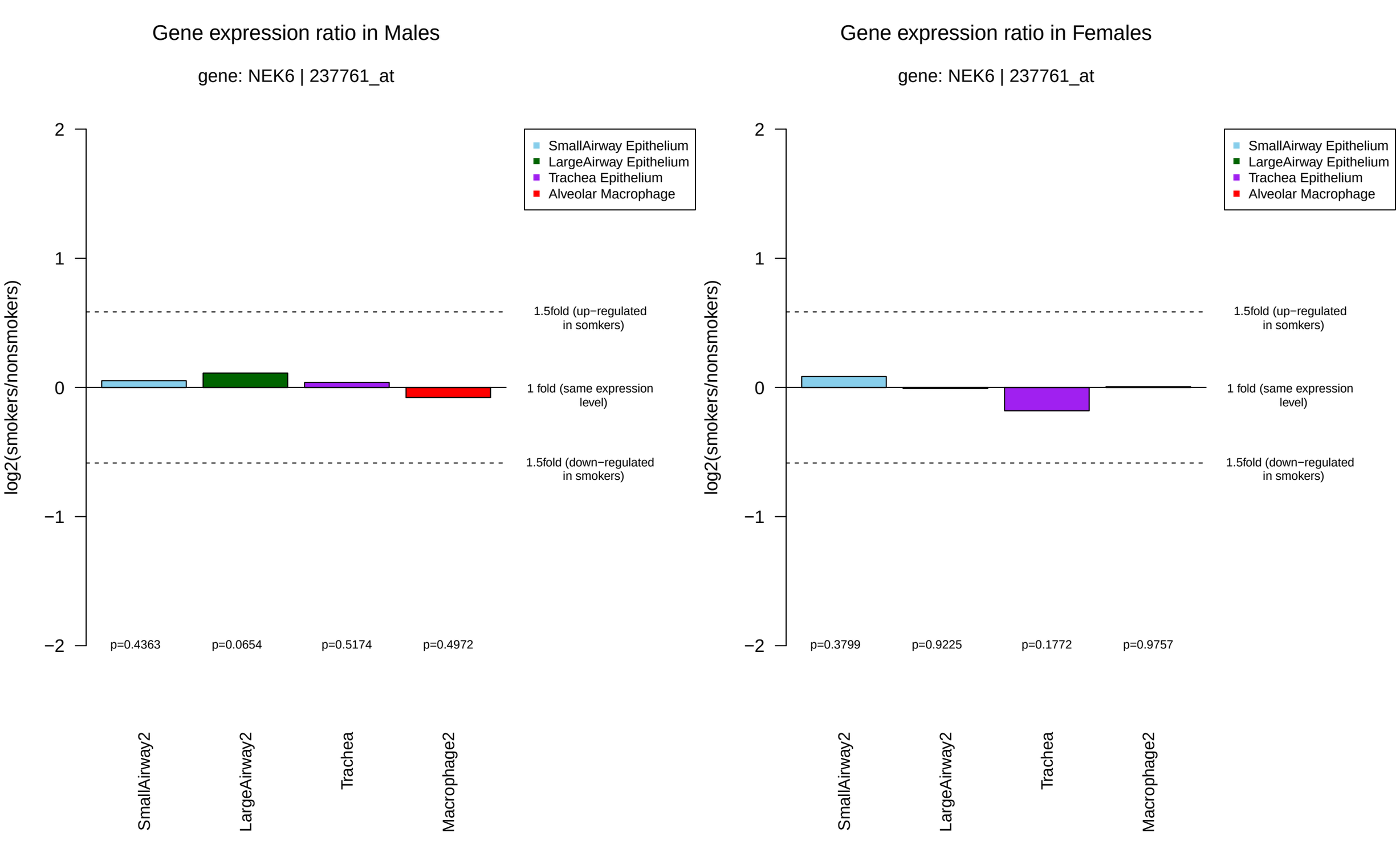
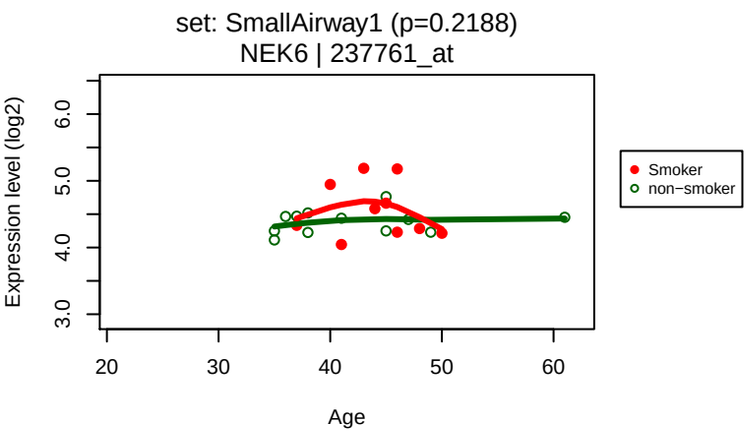
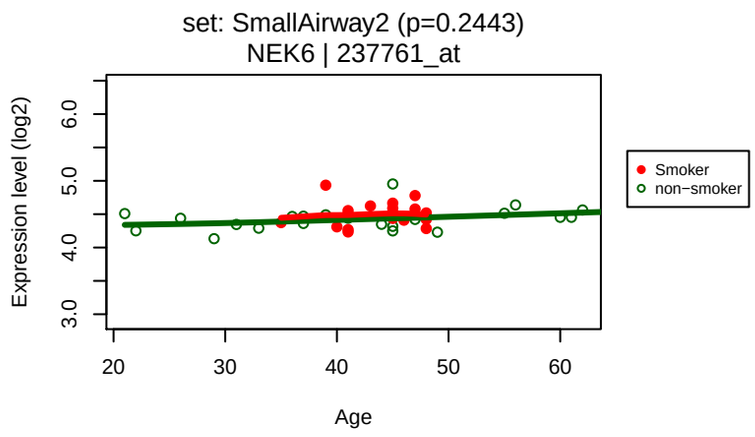
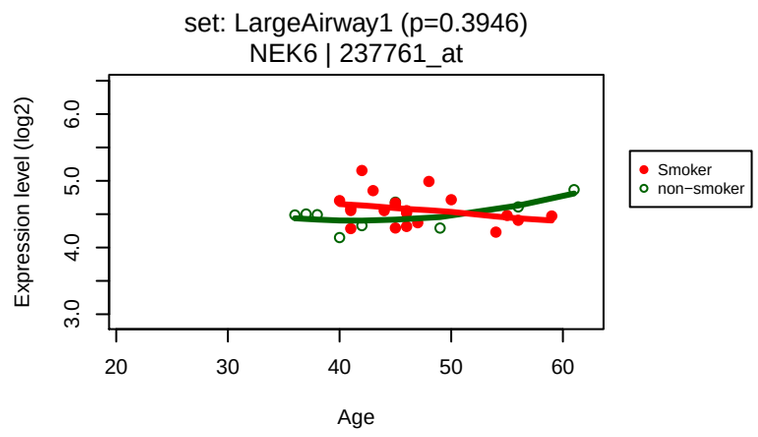

Gene expression of smokers and nonsmokers
gene: NEK6 | 237761_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2188
p=0.2443
p=0.4797
p=0.3946
p=0.2603
p=0.8921
p=0.9817
p=0.605
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

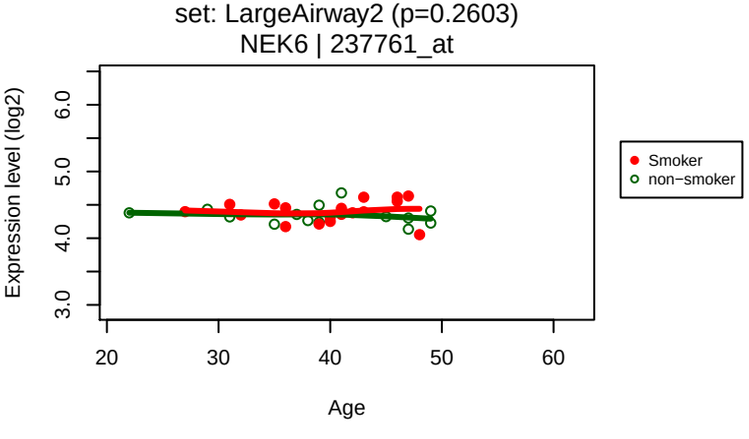

Gene expression ratio (smokers/nonsmokers)
gene: NEK6 | 237761_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2188
p=0.2443
p=0.4797
p=0.3946
p=0.2603
p=0.8921
p=0.9817
p=0.605
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

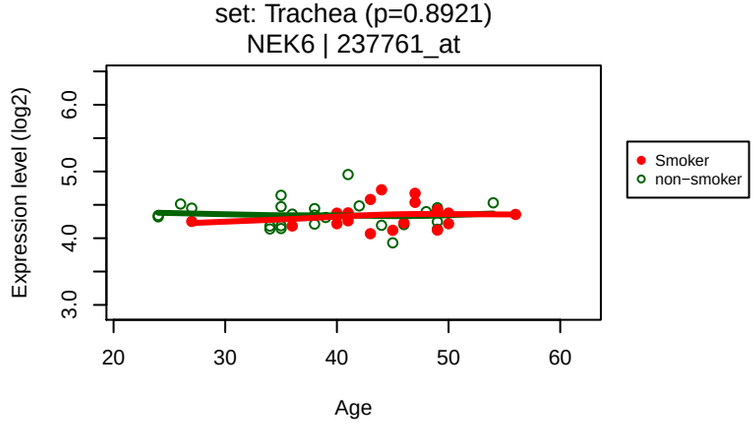

Gene expression ratio in Males
gene: NEK6 | 237761_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4363
p=0.0654
p=0.5174
p=0.4972
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NEK6 | 237761_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3799
p=0.9225
p=0.1772
p=0.9757
SmallAirway2
LargeAirway2
Trachea
Macrophage2

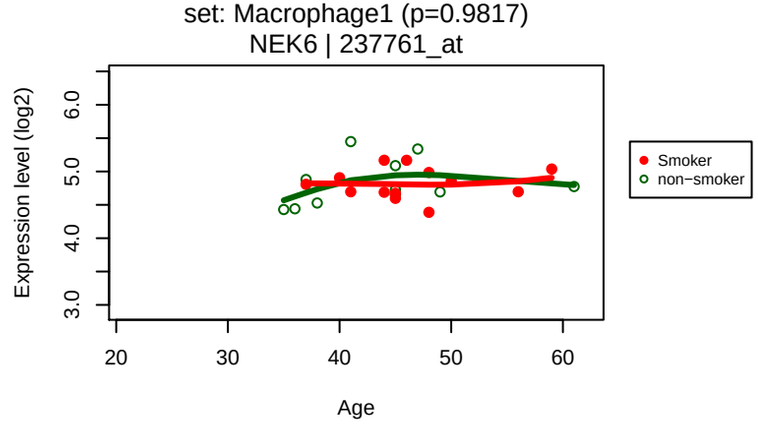

set: SmallAirway1 (p=0.2188)
NEK6 | 237761_at
Expression level (log2)
3.0
4.0
5.0
6.0
20
30
40
50
60
Age
Smoker
non−smoker

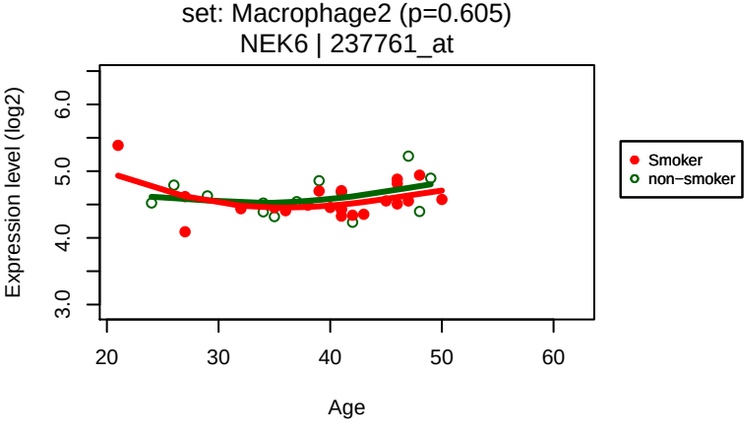

set: SmallAirway2 (p=0.2443)
NEK6 | 237761_at
Expression level (log2)
3.0
4.0
5.0
6.0
20
30
40
50
60
Age
Smoker
non−smoker

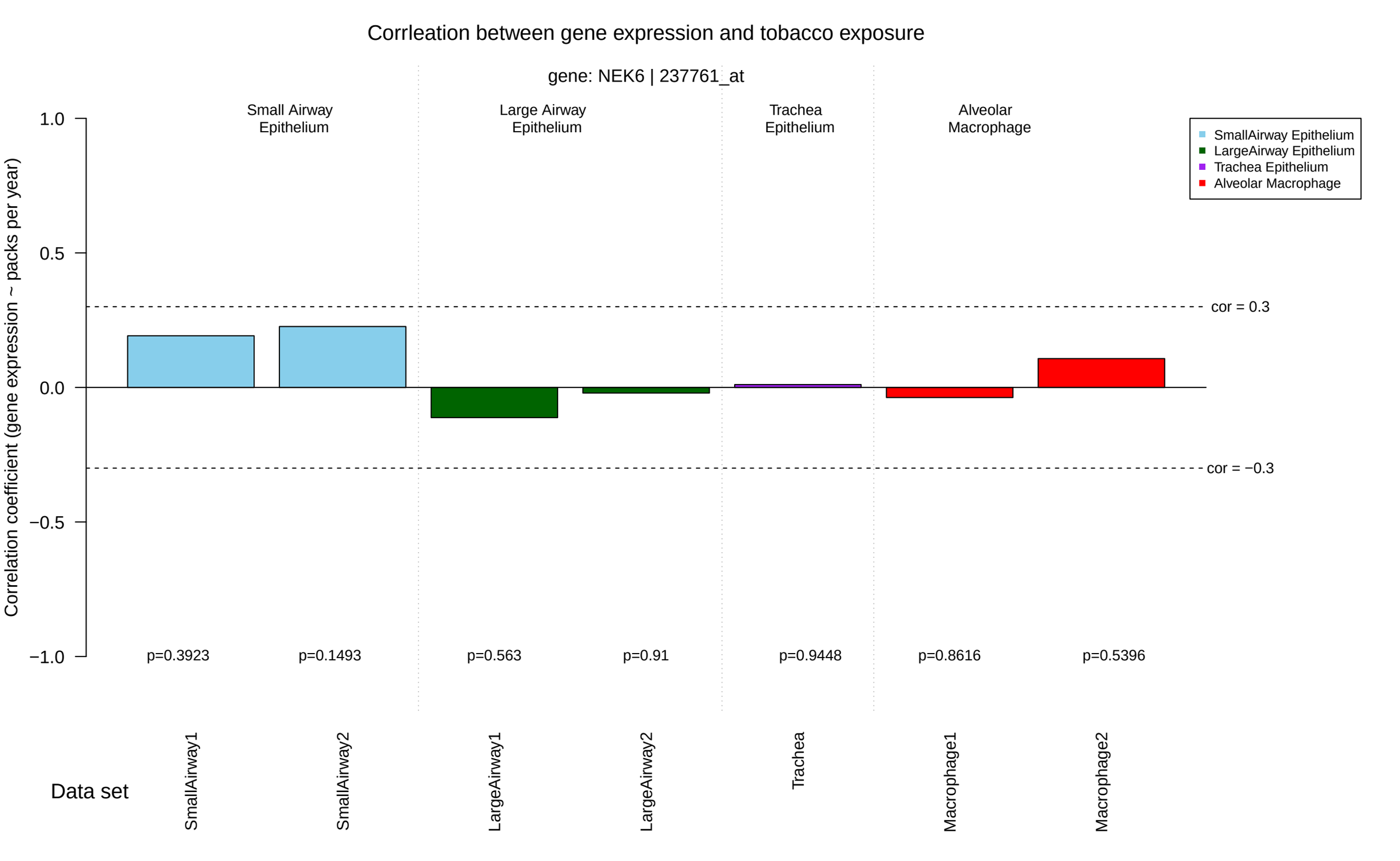

set: LargeAirway1 (p=0.3946)
NEK6 | 237761_at
Expression level (log2)
3.0
4.0
5.0
6.0
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2603)
NEK6 | 237761_at
Expression level (log2)
3.0
4.0
5.0
6.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.8921)
NEK6 | 237761_at
Expression level (log2)
3.0
4.0
5.0
6.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9817)
NEK6 | 237761_at
Expression level (log2)
3.0
4.0
5.0
6.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.605)
NEK6 | 237761_at
Expression level (log2)
3.0
4.0
5.0
6.0
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NEK6 | 237761_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3923
p=0.1493
p=0.563
p=0.91
p=0.9448
p=0.8616
p=0.5396
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2